



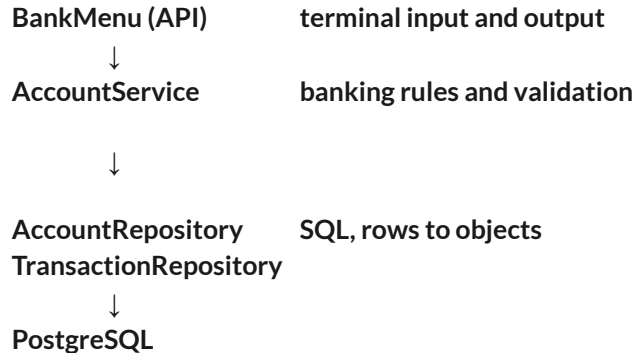
Bank of CLI

Paribesh Upreti

Java · PostgreSQL · JDBC · Docker



Multi-Layered Architecture



- Each layer calls only the one below it.
- The menu writes no SQL.
- The repository makes no decisions.



OOP Supporting the Architecture (APIE)

Abstraction

AccountService calls `findByAccountId()` without knowing about connections, PreparedStatements, or ResultSets. Each layer hides how it works from the one above.

Polymorphism

Connection, PreparedStatement, and ResultSet are JDBC interfaces. The Postgres driver supplies implementations - swap the driver and the code doesn't change.

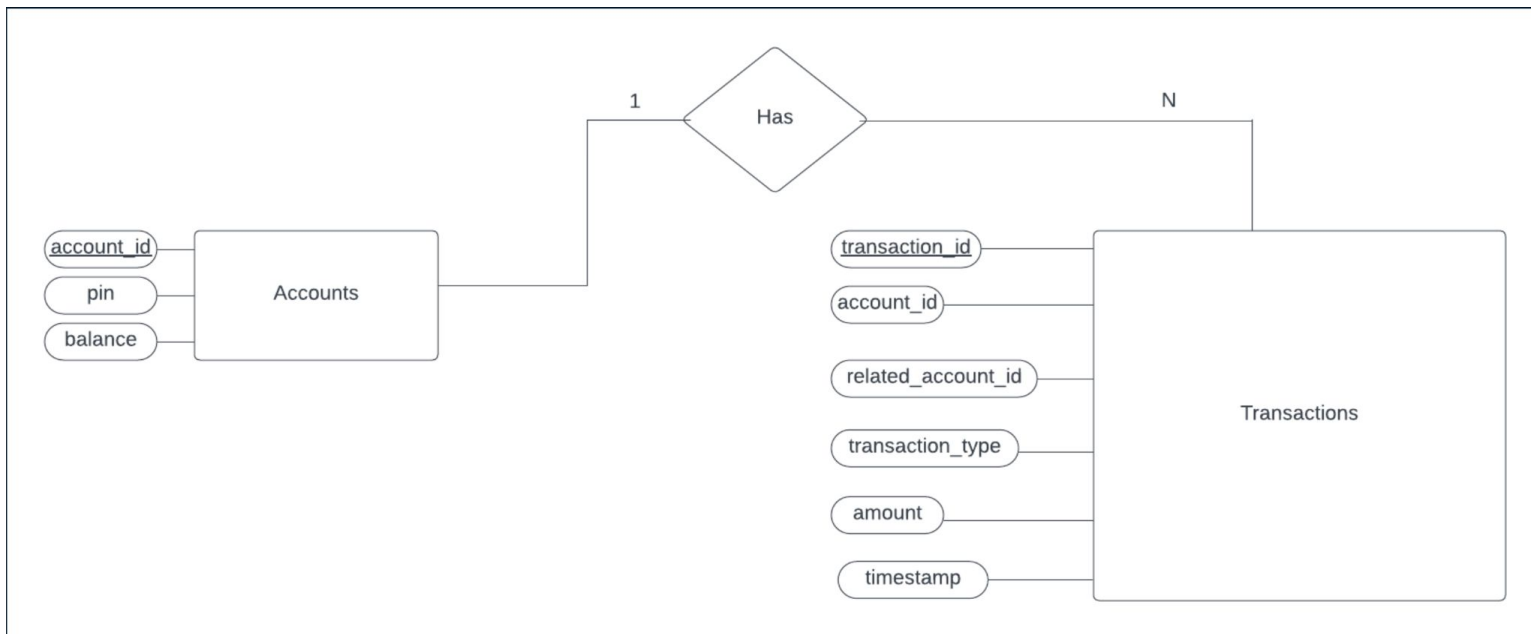
Inheritance

Not used. The design favors composition - AccountService has repositories rather than extending anything.

Encapsulation

Account and Transaction use private fields with getters and setters, so each class controls its own state.

Entity Relationship Diagram





Tech Stack

Java 17 - application language

Maven - build and dependency management

PostgreSQL - two tables, accounts and transactions

JDBC - PreparedStatement, which also prevent SQL injection

JUnit5 - 27 tests, a positive and negative per method

jBcrypt - PIN hashing

Docker - database starts with one command via docker compose

Git / Github - version control



Live Demo